

A Take on Credence, Belief and Sleeping Beauty

based on John MacFarlane's recent paper "Belief: What is it Good for?"

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Premises

- Methodological behaviourism about credence and belief.
- "Don't be Dutch-bookable" as the master norm concerning action governed by decision theory.
- "If you asserted p yesterday, and had good reasons for it, and you have not become aware of any new reasons bearing on the truth of p , then you are still committed to p " as a norm of the language game of giving and asking for reasons.

MacFarlane's Account

- He asks: Which talents do the concepts of "credence" and "belief", respectively, bring to the table, and are both concepts' talents needed when we theorise human beings as rational and social beings?
 - **Credence**: Good at decision-theoretically rationalising and explaining an individual's actions. Belief is much worse at this.
 - **Belief** brings to the table
 - * "truth" as a correctness condition and
 - * being required for the ascription of certain reactive attitudes (cf. Lara Buchak: you should not, e.g., resent someone for stealing your phone if you don't believe they stole your phone).
- What do these talents qualify "belief" for?
 - MacFarlane: For describing and explaining the social practice of *reasoning*. Reasoning proceeds by argument, which works by inferring the truth of a conclusion from the truth of a small set of premises. Furthermore, what e.g. resenting someone requires is exactly a *reason* for resenting them.
- So: we ascribe **belief** that p to someone iff p is a candidate reason for them; if they would use p as a reason in the social practice of reasoning.
- And we ascribe **credence** c in p to someone iff it (decision-theoretically) explains and rationalises their actions.
- Relationship between reasoning and rationality on this account: Very tenuous.
 - Reasoning is about producing, assessing and criticising reasons.
 - Rationality is about the (synchronic and diachronic) coherence of one's credences.

"One can be rational without ever reasoning [...]. Conversely, reasoning—even good reasoning—does not necessarily make one rational."

- MacFarlane still talks about both “Bayesian epistemology” and “practical reasoning”, but I want to radicalize his ideas in a way that would lead us to reject both terms (at least in the individual case) as somewhat misleading. I want to try out the following thesis:
 - The internal states of belief and credence inherit the normative status of the behaviours they cause.
 - Credence ascription is only for explaining an individual’s actions via decision theory. A set of credences is judged by the actions it would cause in this way. Paradigmatically, it is irrational if it would cause one to fall for a Dutch Book. There are (at base level) no epistemic requirements on one’s credences.
 - Belief ascription is only for explaining what one would accept as a reason.
 - * In particular, belief must give credence free reign in an individual’s decision-making.
 - * Where both doing *A* and refraining from *A* is permitted for someone, their act of doing *A* is never accurately described or explained as the result of practical reasoning.
 - * What is normatively required of one’s beliefs is derived from what is required from one’s behaviour in the social practice of (theoretical) reasoning.

Open question: They cannot be totally separate, of course. What is the “pineal gland” that brings them into conversation with one another?

Application to Sleeping Beauty

- The Sleeping Beauty scenario
 - SB is woken up on Monday and Tuesday (with memory loss potion administered in between) if coin lands TAILS
 - SB is woken up only on Monday if coin lands HEADS
 - Upon being woken up, SB is asked: What is the probability of HEADS?
- Applying the above framework:
 - SB’s credence in HEADS has to be $1/3$
 - Why? Dutch Book due to Hitchcock (2004):

Bets	Payoff	Cost	Heads	Tails
bet #1	30 if tails	15	-15	15
bet #2	20 if heads	10	10	-10
bet #3 (only placed if tails)	20 if heads	10 if tails	0	-10
Combined	20 if heads	25 if heads	-5	-5
	30 if tails	35 if tails		

- SB’s belief has to be *that* the probability of *HEADS* is $1/2$: She believes this on Sunday (for good reasons) and has no reason for switching to any other belief.
- (Darren) Bradley and Leitgeb (2006) take this Dutch Book to show that betting odds and credences come apart in *Sleeping Beauty*.
 - This is another possible conclusion we can draw from it.
 - What I take it to, in any case, show is: In *Sleeping Beauty*, we have to sever either the link between credence and truth, or between credence and decision theory.
 - It seems clear that to a MacFarlanian, only the first option is available. But this means giving up the epistemological significance of credence.